

Karl Friston: Neuroscience and the Free Energy Principle | Lex Fridman Podcast #99

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the following is a conversation with Karl Friston one of the greatest neuro scientists in history cited over 245 thousand times known for many influential ideas in brain imaging neuroscience and theoretical neurobiology including especially the fascinating idea of the free energy principle for action and perception Karl's mix of humor brilliance and kindness to me are inspiring and captivating this was a huge honor and a pleasure this is the artificial intelligence podcast if you enjoy it subscribe on youtube review it with five stars in a podcast supported on patreon or simply connect with me on Twitter Alex Friedman spelled Fri D ma n as usual I'll do a few minutes of ads now and never any ads in the middle that can break the flow of the conversation I hope that works for you and doesn't hurt the listening experience this show is presented by cash app the number one finance app in the App Store when you get it used called Lex podcast cash app lets you send money to friends buy Bitcoin and invest in the stock market with as little as \$1 since cash app allows you to send and receive money digitally let me mention a surprising fact related to physical money of all the currency in the world roughly eight percent of it is actual physical money the other 92 percent of money only exists digitally so again if you get cash out from the App Store Google Play and use the code Lex podcast you get ten dollars in cash shop will also donate ten dollars the first an organization that is helping to advanced robotics at STEM education for young people around the world and now here's my conversation with Carl Fuster --an how much of the human brain do we understand from the low level of neuronal communication to the functional level to the to the highest level maybe the the psychiatric disorder level well we're certainly in a better position than we were last century how far we've got to go I think is almost an unanswerable question so you'd have to set the parameters you know what constitutes understanding what level of understanding do you want I think we've made enormous progress in terms of broad-brush principles whether that affords a detailed cartography of the functional anatomy of the brain and what he doesn't

write down to the microcircuitry in the neurons that's probably out of reach at the present time so the cartography so mapping the brain do you think mapping of the brain the detailed perfect imaging of it does that get us closer to understanding of the mind of the brain so how far does it get us if we have the perfect cartography of the brain I think there are lower bounds on that it's a really interesting question you and it would determine this sort of scientific career you'd pursue if you believe that knowing every dendritic connection every sort of microscopic synaptic structure and right down to the molecular level was gonna give you the right kind of information to understand the computational Natale then you choose to be microscopic and you would study little cubic millimeters of brain for the rest of your life if on the other hand you were interested in holistic functions and a sort of functional anatomy of the sort that a neuropsychologist would understand you'd study brain lesions and strokes you know just looking at the whole person so again it comes back to I won't level do you want understanding I think there are principled reasons not to go too far if you commit to a view of the brain as a machine that's performing a form of inference and representing things there are the understanding that level our understanding is necessarily cast in terms of probability densities and ensemble densities distributions and what that tells you is that you don't really want to look at the atoms to understand the thermodynamics of probabilistic descriptions for how the brain works so I personally wouldn't look at the molecules or indeed the single neurons in the same way if I wanted and understand the thermodynamics of some non equilibrium steady state of a gas or an active material I wouldn't spend my life looking at the the individual molecules that constituted there on somebody look at their collective behavior on the other hand if you go to coarse grain you're gonna miss some basic canonical principles of connectivity and architectures I'm thinking here this bitkha local but this current excitement about high field magnetic resonance imaging and seven tests that why well it gives us for the first time the opportunity to look at the brain in action at the level of a few millimeters that distinguish between different layers of the cortex that may be very important in terms of evincing generic principles of canonical microcircuitry that are replicated throughout the brain there may tell us something fundamental about message passing in the brain and these density dynamics of on your own ensemble population dynamics that underwrite our you know our brain function so somewhere between a millimeter and a meter lingering for a bit under and the big questions if you allow me what to you is the most beautiful or surprising characteristic of the human brain I think it's hierarchical and recursive aspect is recurrent aspect of the structure or of the actual representation of power of the brain well I think one speaks to the

other I was actually answering in a more minded way from the point of view of purely its anatomy and its structural aspects I mean there are many marvelous organs in them in the body let's take your liver for example you know without it you wouldn't be around for very long and it does some beautiful delicate chemistry and homeostasis and you're evolved with a finesse that would easily parallel the brain but it doesn't have a beautiful anatomy he has a simple anatomy which is attractive in a minimalist sense but it doesn't have that crafted structure of sparse connectivity and that recurrence and that specialization that the brain has so you said a lot of interesting terms here so the recurrence the sparsity but you also started by saying hierarchical mm-hmm so I've I've never thought of our brain as hierarchical sort of I always thought is just like a giant mess an interconnected mess was very difficult to figure anything out but in what sense do you see the brain is hierarchical well I see it's not a magic soup yeah of course it's what I used to think when I was before I studied medicine and the like so a lot of those terms imply each other so hierarchies if you just think about the nature of a hierarchy how would you actually build one and what you would have to do is basically carefully remove the right connections that destroy the completely connected soups that you might have in mind so a hierarchy is in and of itself defined by a sparse and particular connectivity structure I'm not committing to any particular form of hierarchy the your senses there is some oh absolutely in virtue of the fact that there is a sparsity of connectivity not necessarily of a quality it's obvious and if a quantitative sort so they are it is demonstrably so and that they're far further apart two parts of the brain are the less likely that they are to be wired you know to possess axonal processes neuronal processes that directly communicate one message or messages from one part of the brain to the other part of the brain so we know there's a sparse connectivity and furthermore on the basis of anatomical connectivity and traces studies we know that that has that sparsity underwrites a higher high rock on a very structured sort of connectivity that might be best understood like a little bit like an onion you know that there there is a concentric sometimes refer to as centripetal by people like Marcel messulam hierarchical organization to the brain so you can think of the brain as in a rough sense like an onion and all the sensory information and all the afferent outgoing messages that supply commands to your muscles or to your secretory organs come from the surface so there's a massive exchange interface with the world out there on the surface and then underneath there's a little layer that sits and looks at the exchange on the surface and then underneath that there's a layer right there way down to the very center through the deepest part of the onion that's what I mean by a mirror hierarchical

organization there's a discernible structure defined by the sparsity of connections that lends the architecture a hierarchical structure that tells one a lot about the kinds of representations and messages so far on any question is this about the representational capacity or is it about the anatomy well one under writes the other you know if one this simply thinks of the brain as a message passing machine a process that is in the service of doing something then the the circuitry and the connectivity that shape that message passing also dictate its function so you've done a lot of amazing work in a lot of directions so let's look at one aspect of that of looking into the brain and trying to study this onion structure of what can we learn about the brain by imaging it which is one way to sort of look at the anatomy of it broadly speaking what what are the methods of imaging but even bigger what can we learn about it right so well most imaging human neural imaging you might see you know in science journals the speaks to the way the brain works measures brain activity over time so you know that's the first thing to say the way we're effectively looking at fluctuations in neuronal responses usually in response to some sensory input or some instruction some task not necessarily and there's a lot of interest in just looking at the brain in terms of resting state endogenous or intrinsic activity but crucially at every point looking at these fluctuations either induced or intrinsic in the neural activity and understanding them at two levels so normally people would recourse to two principles of brain or kin organization that are complimentary one functional specialization or segregation so what does that mean it simply means that there are certain parts of the brain that may be specialized for certain kinds of processing you know for example visual motion our ability to recognize or to perceive movement in the visual world and furthermore that specialized processing may be spatially or anatomically segregated leading to functional segregation which means that if I were to compare your brain activity during a period of studying viewing a static image and then compare that to the responses of fluctuations in the brain when you are exposed to a moving image say a flying bird eirick we would expect to see restricted segregated differences in activity and those are basically the hot spots that you see in me in surgical parametric maps that test for the significance of the responses that are circumscribed so now basically we're talking about some people of perhaps and currently Calder and neocartography this is a phrenology augmented by modern day near imaging basically finding blobs or bumps on the brain that do this or do that and trying to understand the cartography of that functional specialization so how much how much is there such this is such a beautiful sort of ideal to strive for we we humans scientists would like you like this to hope that there is a beautiful structure to this was like you said there are segregated regions

that are responsible for the different function how much hope is there to find such regions in terms of looking at the progress of studying the brain oh I think in Nomis progress has been made in the past you know 20 or 30 years you know so this is beyond incremental you know at the advent of brain imaging the very notion of functional segregation was just a hypothesis based upon a century if not more of careful neuropsychology looking at people who had lost via insult or traumatic brain injury particular parts of the brain and then saying well they can't do this or they can't do that for example losing the visual cortex and not being able to see or using losing particular parts of the visual cortex or regions known as v5 or the middle temporal region MT noticing that they selectively could not see moving things and so that created the the hypothesis that perhaps movement processing visual movement processing was located in this functionally segregated area and you could then put go and put invasive electrodes in animal models and say yes indeed we can excite activity here we can form receptive fields that are sensitive to or defined in terms of visual motion but at no point could you exclu the possibility that everywhere else in the brain was also very interested in visual motion by the way I apologize to interrupt buzz tiny little tangent you said animal models just out of curiosity from your perspective how different is the human brain versus the other animals in terms of our ability to study the brain well clearly the far further away you go from a human brain the the greater the difference is but not not as remarkable as you might think so people will choose their level of approximation to the human brain depending upon the other kinds of questions that they want to answer so if you're talking about sort of canonical principles of microcircuitry it might be perfectly okay to look at a mouse indeed you could even look at flies worms if on the other hand you wanted to look at the finer details of organization of visual cortex and v1 v2 there's a designated sort of patches of cortex that may or may do different things indeed do you probably want to use a primate that looked a little bit more like a human because there are lots of ethical issues in terms of you know the use of non-human primates to transfer questions about the about human anatomy I think most people assume that the most of the important principles are conserved in a continuous way you know from right from well yes worms right to yummy so now returning to so that was the early of ideas are studying the the the really functional regions of the brain base if there's some damage to it to try to infer that there's that part of the brain might be somewhat responsible for this type of function so what where does that lead us what are the next steps beyond that right well this actually reverse a bit come back to your sort of notion that the brain is a magic sue but that was actually a very prominent idea at one point notions such as Lashley's law of mass action

inherited from the observation that for some animals if you just took out spoonfuls of the brain it didn't matter where you took these spoonfuls out they always showed the same kinds of deficits so you know it was very difficult to infer functional specialization pure on the basis of lesion deficit studies but once we had the opportunity to look on the brain or lighting up in it's literally it's sort of excitement neuronal AM excitement when looking at this versus that one was able to say yes indeed these functionally specialized responses are very restricted and then they're here or they're over there if I do this then this part of the brain lights up and that became doable in the early 90s in fact shortly before with the advent of positron emission tomography and then functional magnetic resonance imaging came along in the early 90s and since that time there has been an explosion of discovery refinement confirmation you know there are people who believe that it's all in the anatomy if you understand the anatomy then you understand the function at some level and many many hypotheses were predicated on a deep understanding of the anatomy and the connectivity but they were all confirmed and taking much further with newer imaging so that's what I meant by we've made an enormous amount of progress in in this century indeed and in relation to the previous century by looking at these funky selective responses but that wasn't the whole story so there's a sort of near phrenology but finding bumps and hotspots in the brain that did this or that the bigger question was of course the functional integration how all of these regionally specific responses were orchestrated how they were distributed how did they relate to distributed processing and indeed representations in the brain so then you turn to the more challenging issue of the integration the connectivity and then we come back to this beautiful sparse recurrent hierarchical connectivity that seems characteristic of the brain and probably not many other organs and but nevertheless we'll come back to this this challenge of trying to figure out how everything is integrated but what's your feeling what's the general consensus how we moved away from the magic soup view of the brain yes so there is a deep structure to it yeah that and then maybe further question you said some people believe that the structure is most of it that you can really get at the core of the function by just deeply understanding the structure yeah where do you sit on that do you I think it's called some monster yes yeah yes it's a worthy pursuit of going of studying of through imaging and all the different methods to actually study no absolutely let's go yeah yeah sorry I'm just I'm just nutty you you were accusing me of using lots of long words and then you introduce one that which is deep which is interesting and because deep is this or Millennial equivalent of hierarchical so if you've put a deep in front of anything you're very millennial and start trending but you yes you're also implying a

hierarchical architecture so that's it is a depth which is for me the beautiful thing that's right the word deep kind of yeah exactly it implies hierarchy I didn't even think about that that indeed the implicit meaning of the word deep is a hierarchy yep yeah yeah so deep inside the onion is a central view so if you put maybe briefly if you could paint a picture of the kind of methods of neuro imaging maybe the history which you are a part of you know from statistical parametric mapping I mean just what what's out there that's interesting for people maybe outside the field that to understand of what are the actual methodologies of looking inside the human brain right well there you can answer that question from two perspectives basically it's the modality you know what kind of signal are you measuring and they can range from and let's limit ourselves to some imaging based non-invasive techniques so you've essentially got brain scanners and Brent's cannons can either measure the structural attributes the amount of water of the Mount of fat on the amount of iron in different parts of the brain you can make lots of inferences about the structure of the organ of the sort that you might have abuse from an x-ray but a you know a very nuanced x-ray that is looking at this kind of property of that kind of property so looking at the anatomy not invasively is would be the first sort of earner imaging that people might want to employ then you move on to the kinds of measurements that reflect dynamic function the most prevalent of those fall into two camps you've got these metabolic sometimes hemodynamic blood related signals so these metabolic and/or hemodynamic signals are basic proxies for elevated activity and message passing and neuronal dynamics in particular parts of the brain characteristically though the time constants of these hemodynamic or metabolic responses to neural activity are much longer than the neural activity itself and this is uh this is refering forgive me for the dumb questions but this would be referring to blood like the flow of blood absolutely so there's a ton of it seems like there's a ton of blood vessels in the brain yeah so but what's the interaction between the flow of blood and the function of the new and neurons is there an interplay there or yeah yeah yeah and that interplay accounts for several careers of world-renown solutely so this is known as neurovascular coupling is exactly what you said it's how how does a neural activity the neuronal infrastructure natural message passing that we think underlies our capacity to perceive and act how is that coupled to the vascular responses that that supply the energy for that neural processing so there's a delicate web or of large vessels arteries and veins that gets progressively finer and finer in detail until it perfuses at a microscopic level the machinery where little neurons lie so coming back to this sort of onion perspective we were talking before using the onion there's a metaphor for a deep hierarchical structure but also I think it's

just an anatomical metaphor quite a useful metaphor all the action all the heavy lifting in terms neural computation is done on the surface of the brain and then the interior of the brain is constituted by fatty wires essentially axonal processes that are enshrouded by myelin sheaths and these give the ER when you dissect them they look fatty and white and so it's called white matter as opposed to the actual neuro peel which does the computation constituted largely by neurons and that's known as gray matter so the gray matter is a a surface or a skin that sits on top of this big ball now we are talking magic soup but it's a big ball of collections like spaghetti very carefully structured with sparse connectivity that preserve this deep hierarchical structure but all the action takes place on the surface on the cortex of the onion and that means that you have to supply the right amount of blood flow the right amount of nutrient which is rapidly absorbed and used by neural cells that don't have the same capacity that your leg muscles would have to basically spend their energy budget and then claim it back later so one peculiar thing about cerebral metabolism brain metabolism is it really needs to be driven in the moment which means you basically have to turn on the taps so if there's lots of neural activity in one part of the brain a little patch of a couple of millimeters even less possibly you really do have to water that piece of the garden now and quickly and that by quickly I mean within a couple of seconds so that contains a lot of information the imaging could tell you a story of what's happening absolutely but it is slightly compromised in terms of the resolution so the the deployment of these little micro vessels that the water the garden to enable the activity to to the neural activity to play out the the spatial resolution is in order of a few millimeters and crucially the temporal resolution is the order of a few seconds so you can't get right down and dirty into the actual spatial and temporal scale of neuronal activity in and of itself to do that you'd have to turn to the other big imaging modality which is the recording of electromagnetic signals as they're generated in real time so here the temporal bandwidth if you like on the temp the low limit on the temporal resolution is incredibly small you're talking about near nalle' seconds milliseconds and then you can get into the phasic fast responses there is in of itself the neural activity and start to see the succession or cascade of hierarchical recurrent message-passing evoked by a particular stimulus but the problem is you're looking at electromagnetic signals that have passed through an enormous amount of magic soup or spaghetti of collectivity and through the scalp and the skull and it's become spatially very diffused so it's very difficult to know where you are so you've got this sort of catch-22 you can either use an imaging modality it tells you within millimeters which part of the brain is activated we don't know when or you've got these electromagnetic a EEG m EG setups that

tell you to within a few milliseconds when folks something has responded being aware so you've got these two complementary measures either in direct via the blood flow or direct via the electromagnetic signals caused by neural activity these are the two big imaging devices and the second level of responses your question what what are they yeah from the outside one of the big ways of using this technology so once you've chosen your the kind of mirror imaging they want to use to answer your set questions and sometimes it would have to be both then you've got a whole raft of analyses time series analysis usually that you can bring to bear in order to answer your questions or address your hypothesis about those data and interesting that they they've both fall into the same two camps we're talking about before you know this dialectic between specialization and integration differentiation and integration so it's the cartography that blob ology analyses my apology and probably shouldn't transfer much but just the herd of fun word the blur the robot ology blood ology its ideologies of which means the study of blobs that's nothing for are you being witty and humorous or is there an actual there's the word blob ology ever appear in a text book somewhere it would appear in a popular book it would not appear in a worthy specialist journal yeah it's the fond word for the study of literally little blobs on brain maps showing activations so the kind of thing that you'd see in you know the newspapers on ABC or BBC reporting the latest finding from a from brain imaging interestingly though the maths involved in that stream of analysis does actually call upon the mathematics of blobs so seriously they actually called Euler characteristics and you know they have a lot of fancy names in mathematics we'll talk about about your ideas in free energy principle I mean there's a echoes of blobs there when you consider sort of entities so mathematically speaking yes absolutely yeah yes anyway well the first circumscribe well-defined yes--you entities of well in from the free energy point of view entities of anything but from the point of view of the analysis the cartography of you know of the brain these are the entities that constitute the evidence for this functional segregation you have segregated this function in this blob alledge is not outside of the blob that's basically the oh if you were a map maker of America and you did not know instruction the first thing were you doing constituting or creating a map will be to identify the cities for example or the route mountains and all the rivers all of these uniquely spatially localizable features possibly topological features have to be placed somewhere because that requires our mathematics of identify what does a set it City look like on a satellite image or what does a river look like I want as a mountain look like what would it you know what data features wood is wood evidence that that particular table you know that particular thing that you wanted to put on the map and they normally are

characterized in terms of literally these blobs or these of now the way looking at and this is a certain statistical measure of the degree of activation crosses a threshold and in crossing that threshold in a spatially restricted part of the brain it creates a blob and that's basically what physical parametric mapping does it's basically mathematically finessed phlebology okay so those you kind of describe these two methodologies for one is temporally noisy one is spatially noisy and you kind of have to play and figure out what what can be useful yeah it'd be great if you can sort of comment I got a chance recently to spend a day at a company called neural link that uses brain computer interfaces and their dream is to well there's a bunch of sort of dreams but one of them is to understand the brain by sort of you know getting in there past the so calls that are factory wall getting in there be able to listen communicate both directions what are your about this the future of this kind of technology of brain computer interfaces to be able to now have a have a window or direct contact within the brain to be able to measure some of the signals to be able to send signals to understand some of the functionality of the brain ambivalent my sense is ambivalent so it's a mixture of good and bad and I acknowledge that freely so the good bits if you just look at the legacy of that kind of reciprocal but invasive geo brain stimulation I didn't paint a complete picture when I was talking about some of the ways we understand the brain prior to your imaging it wasn't just leave lesion deficit studies some of the early work in fact literally a hundred years from where we're sitting at the institution neurology what was done by stimulating the brain of say dogs and looking at how they responded either but with them the muscles or with the salivation and imputing what that part of the brain must be doing that if i stimulated then yeah and i vote this kind of response then that tells me quite a lot about the functional specialization so there's a long history of brain stimulation which in continues to enjoy a lot of attention nowadays positive attention oh yes absolutely you know deep brain stimulation for Parkinson's disease is now a standard treatment and also a wonderful vehicle to try and understand the neuronal dynamics underlie movement disorders like Parkinson's disease even interest in transmitting its magnetic stimulation stimulating with the magnetic fields and will it work in people who depressed for example quite a crude level of understanding what you're doing but you know there are there is historical evidence that these kinds of brute force and interventions do change things then you know a little bit like buying the TV whether the valves are working properly but it still it works so you know there is a long history brain computer interfacing a BCI I think is a beautiful example of that it's sort of carved out its own lesion its own aspirations and they've been enormous advances within limits advances in terms of our ability to understand how the

brain the embodied brain engages with the world I'm thinking of here of sensory substitution the augmenting our sensory capacities by giving ourselves extra ways of sensing than sampling the world ranging from sort of trying to replace lost visual signals through to giving people completely new signals so the well I think most engaging examples of this is equipping people with a sense of magnetic fields so you can actually give them magnetic sensors that enable them to feel should we say tactile pressure around their tummy where they are in relation to them to the magnetic field of the earth incredible and after a few weeks they take it for granted they integrated the embody assimilate this new sensory information into the way that they feel literally feel their world were now equipped with this sense of magnetic direction so that tells you something about the brain's plastic potential to remodel to in term and its plastic capacity to suddenly try to explain the sensory data at hand by augmenting or augmenting the the sensory sphere and the kinds of things that you can measure clearly that's purely for entertainment and understanding the knee or the nature and the power of our brains I would imagine the most BCI is pitched at solving clinical and human problems such as locked-in syndrome paraplegia or replacing lost sensory capacitors like blindness and death deafness so then we come to the more on the negative part of my own the other side of it so I you know I don't want to be deflation because much of my deflationary comments was probably large out of ignorance there anything else but generally speaking the the bandwidth and the bit rates that you get from brink of Pewter interfaces as we currently know them we're talking about bits per second so that would be like me only being able to communicate with any world or with you using very very very slow Morse code and it is not in the in even within an order of magnitude near what we actually need for an inactive realization of what people aspire to when they think about sort of curing people with paraplegia or replacing site despite heroic efforts so one has to ask is there a lower bound on the kinds of recurrent information exchange between a brain and some augmented or artificial interface and let me come back to interestingly what I was talking about before which is your if you're talking about function in terms of inference and I presume we'll get to that later on in terms of the free energy principle Minh the moment they may be fundamental reasons to assume that is the case we talk about ensemble activity we're talking about basically for example let's paint challenge facing brain-computer of interfacing in terms of controlling another system that is highly and deeply structured very relevant to our lives very nonlinear the rests upon the kind of non-equilibrium steady states and dynamics that the brain does the weather right so good example here imagine you had some very aggressive satellites that could produce signals that could be termed some

little parts of the of the weather system and then what you're asking now is can i meaningfully get into the weather and change it meaningfully and make the weather respond in a way that I want it to you're talking about chaos control on a scale which is almost unimaginable so there may be fundamental reasons why BCI as you might read about it in a science fiction novel aspirational BCI may never actually work in the sense that to really be integrated and be part of the system isn't impermanent requires you to have evolved with that system that you know you you have to be part of a very delicately structured deeply structured dynamic ensemble activity that is not like rewiring a broken computer or plugging in a peripheral interface adapter it is much more like getting into the weather pans or a come back to your magic soup is getting into the active matter and meaningfully relate that to the outside world so I think there are an enormous challenges there so I think the example the weather is a brilliant one and I think you paint a really interesting picture and it wasn't as negative as they thought it's essentially saying there's it might be incredibly challenging including the lower bound of the bandwidth and so on I kind of so and just to full disclosure I come from the machine learning world so my my natural thought is the hardest part is the engineering challenge of controlling the weather of getting those satellites up and running in and so on and once they are then the rest is of fundamentally the same approaches that allow you to be to win in the game of Go will allow you to potentially play in this soup in this chaos so I have I have a hope that so machine learning methods will will help us play in the soup as but perhaps you're right that it is a via biology and the brain is just an incredible an incredible system that may be almost impossible to get in but for me what seems impossible is is the incredible mess of blood vessels that you also described without you know we also value the brain you can't make any mistakes you can't damage things so to me that engineering challenge seems nearly impossible one of the things I was really impressed by at neuro-link is just just talking to brilliant neurosurgeons and the roboticists that it made me realize that even though it seems impossible if anyone can do it it's some of these world-class engineers that are trying to take it on so so I think the conclusion of our discussion here is of this part is is basically that the problem is really hard but hopefully not impossible absolutely so if it's ok let's start with the basics so you've also formulated a fascinating principle the free energy principle could we maybe start at the basics and what is the free energy principle well in fact the free energy principle inherits a lot from the building of these data analytic approaches to these you know very high dimensional time soon as you get get from the brain so I think is interesting to acknowledge that and in particular the analysis tools that try to address the other side which is a functional

integrations on the connectivity analyses on the one hand but I should also acknowledge it inherits an awful lot from machine learning as well so the free energy principle and is just a formal statement that the the existential imperatives for any system that manages to survive in a changing world is can be cast as a an inference problem in the sense that you can interpret the probability of existing as the evidence that you exist and if you can write down that problem of existence as a statistical problem that you can use all the maths that has been developed for inference to understand and characterize the ensemble dynamics that must be in play in the service of that inference so technically what that means is you can always interpret anything that exists in virtue or being separate from the environment in which it exists as trying to minimize variational free-energy and if you're from the machine learning community you will know that as a negative evidence lower bound or a negative elbow which is the same as saying you're trying to maximize or it will look as if all your dynamics are trying to maximize the complement of that which is the marginal likelihood or the evidence for your own existence so that's basically that you know that the free energy principle of it but even take a sort of a small step back or as you said the existential imperative there's a lot of beautiful poetic words here but to put it crudely there's a it's a fascinating idea of basically just of trying to describe if you're looking at a blob how do you know this thing is alive what does it mean to be alive what does it mean to be to exist and so you can look at the brain you can look at parts of the brain or you this is just the general principle that applies to almost thing and ye and you system it that's just a fascinating sort of philosophically at every level question and the methodology to try to answer that question what does it mean to be alive yeah so that that that's a huge endeavor and it's nice that there's at least some from some perspective a clean answer so maybe can you talk about that optimization view of it so what what's trying to be minimized to maximize what a system that's alive what is it trying to minimize right you've you've made a big move yes to make big moves but you've assumed that the things the thing exists before the in a state that could be living on nonliving so I may ask you or what licenses you to say that something exists that's why I use the word existential it's beyond living it's just existence so if you drill down onto the definition of things that exist then they have certain properties if you borrow the maths from non-equilibrium steady state physics that enable you to interpret their existence in terms of this optimization procedure so it's good you introduce the word optimization so what the free-energy principle in its sort of most ambitious but also most deflationary and simplest says is if something exists then it must by the mathematics of non-equilibrium steady state exhibit properties that may look as if it is optimizing a particular

quantity and it turns out that particular quantity happens to be exactly the same as the evidence lower bound in machine learning or Bayesian model evidence in Bayesian statistics or and then I can list a whole other you know list of ways of understanding this this this key quantity which is a bound on on surprisal self information if you know information theory there are whole there are a number of different perspectives on this contrary it's this basically the log of probability of being in a particular state I'm telling this story as an honest attempt to answer your question and I'm answering it as if I was pretending to be a physicist who was trying to understand the fundamentals of non-equilibrium steady state and I shouldn't really be doing that because the last time I was taught physics I was in my twenties what kind of systems when you think about the free energy principle what kind of systems are you imagining it's a sort of more specific kind of case study you know I'm imagining a range of systems but you're at its simplest a simple single-celled organism that can be identified from its environment or its environment so at its simplest that that's basically what what I always imagined in my head and you may ask well is there anything how on earth can you even in elaborate questions about the existence of a single drop of oil for example yeah what but there aren't deep questions there why doesn't the oil why doesn't the thing the interface between the drop of oil that contains an interior and the thing that is not the drop of oil which is the solvent in which it is immersed how does that interface persist over time why doesn't the oil dissolve into solvent so what special properties of the exchange between the surface of the oil drop and the external states in which it's immersed if you're physicists say would be the heat bath you know you've got a you've got a physical system an ensemble again with about ten thousands ensemble dynamics an ensemble of thousands of atoms or molecules immersed in the heat bath but the question is how did the heat bath get there and why is it not dissolved why was it maintaining itself exactly what actions is it I mean it's such a fascinating idea of a drop of oil and I guess it would dissolve in water wouldn't dissolve in water so what precisely so why not so why not why not and how do you mathematically describe me is such a beautiful idea and also the idea of like where does the thing where does the drop of oil and yeah and where does it begin right so I mean you're asking the questions deep in in a normal area but what you can do you see so this is a deflationary part of it can I just qualify mouths so by saying that normally when I'm asked this question I answer from the point of view of a psychologist we talk about predictive processing and pretty coding and you know the brain is an inference machine but you haven't asked me from that perspective I'm answering from the point of view of a physicist so you you know the question is not so much why but if it exists

what properties must it display so that's the deflation in part the 300 prints we print the 300 principal does not supply and answer as to why it's saying if something exists then he must display these properties that's that's the other sort of the thing that's on offer and it so happens that these properties a must display are actually intriguing and have this inferential gloss this there's this sort of self evidencing loss that inherits from the fact that the very preservation of the boundary between the oil drop and the not oil drop requires an optimized of a particular function or a functional that's that defines the presence of the existence of of this order which is why I started with existential imperatives and the it ISM it is a necessary condition for existence that this must occur because the thing the boundary basically defines the things that's existing so it is that self-assembly aspect it's that for the you hinting at in biology sometimes known as Auto poiesis in computational chemistry Mis self-assembly it's the what what does it look like sorry how would you describe things that configure themselves out of nothing the where they clearly demarcate themselves from the states on the soup in which they are immersed so from the point of view of computational chemistry for example you just understand that as a configuration of a macromolecule to minimize its free energy is thermodynamic free energy it's exactly the same principle that we've been talking about that thermodynamic free energy is just the negative elbow it's the same mathematical calm construct so the very emergence of existence of structure or form that can be distinguished from the environment or the thing that is not the thing necessitates the you know the existence of an objective function then it looks as if it is minimizing it's finally a free energy minima and so just to clarify I'm trying to wrap my head around so the the free energy principle says that if something exists these are the properties it should display yes so what what that means is we can't just look we can't just go into a soup and there's no mechanism if free energy principle doesn't give us a mechanism to find the things that exist is that what it was implying is being applied that you can kind of use it to reason to think about like study a particular system and say does this exhibit these qualities that's an excellent question to answer that after I have to return to your previous question but what's the difference between living and nonliving things actually Society so yeah that maybe we can go there you kind of drew a line and and forgive me for the stupid questions but the you kind of draw a line between living and existing yeah is there an interesting sort of distinction distinction yeah I think there is so you know things do exist grains of sand rocks on the moon trees you so all of these things can be separated from the environment in which they are immersed and therefore there must at some level be optimizing their free energy taking this sort of model evidence interpretation of this quantity

that basically means their self evidencing another nice little twist of phrase here is that you are your own existence proof you know statistically speaking which I don't think I said that somebody did but I love that phrase you are your own existence proof yeah so it's ur existential isn't it I'm gonna have to think about there for a few days yeah the view there's a beautiful line so the the step through to answer your question about you know what's it good for big girl on the following lines first of all you have to define what it means to exist which down as you rightly pointed out you have to define what probabilistic properties must the states of something possess so that it has so it knows where it finishes and then you write out that down in terms of statistical independence is again sparsity again it's not what's connected or what score elated or what depends upon what it's what's not correlated and what doesn't depend upon something again it comes down to the the deeper structures not in this is hierarchal but the suddenly the the structures that emerge from removing connectivity in dependency in this instance basically being able to identify the surface of the oil drop from the water in which it is immersed and when you do that you start to realize well there are actually four sub kinds of states in any given universe that contains anything the things that are internal to the surface the things that are external to the surface and the surface in and of itself which is why I use a metaphor a little single-celled organism that has an interior and exterior and then the the surface of the cell and that's mathematically a Markov blanket just to pause I'm in awe of this concept that there's the stuff outside the surface stuff inside the surface in the surface itself the Markov blanket it's just the most beautiful kind of notion about trying to explore what it means to exist you're automatically I apologize this is a beautiful idea but came out of California so that's I changed my mind I take it all so sorry anyway so what you were just talking about the surface about the market yeah so this surface or this blanket these blanket states that are this you know the because they are now defined in relation to these Independence's and your Walker what different states internal or blanket or external states can which ones can influence each other and which cannot influence each other you can now apply standard results that you would find in non equilibrium physics or steady state or thermodynamics or hydrodynamics usually out of equilibrium solutions and apply them to this partition and what it looks like as if all the Norman normal gradient flows that you would associate with any non equilibrium system apply in such a way that to part of the Markov blanket and the internal states seem to be hill climbing or doing a gradient descent on the same quantity and that means that you can now describe the very existence of this oil drop you can write down the existence of this holdup in terms of flows dynamics equations of motion where

the blanket States or part of them we call them active States and the internal states now seem to be and must be trying to look as if they're minimizing the same function which is a lot of probability of occupying this but these states the interesting thing is that what would they be called if you were trying to describe these things there were what we're talking about are internal states external states and blanket States now let's carve the blanket States into to sensory states and active States operationally it has to be the case that in order for this carving up in two different sets of states to exist the active States the Markov blanket cannot be influenced by the external states and we already know that the internal States can't be influenced by the external States cousin the blanket separates them so what does that mean well it means the active States the internal states are now jointly not influenced by external states they only have autonomous dynamics so now you've got a picture of an oil drop that has autonomy it has autonomous States it has autonomous days in the sense that there must be some parts of the surface of the oil drop that are not influenced by the external states and all the Interior and together those two states endow even a little oil drop with autonomous states that look as if they are optimizing their variational free energy or their negative entropy their model evidence and that would be an interesting intellectual exercise and you could say you can even go into the realms of panpsychism that everything that exists is implicitly making inferences on self-evidencing now we made the next move but what about living things I mean so let me ask you what's the difference between an oil drop and a little tadpole or a little lava or plankton the picture which is painted of an oil drop just immediately in a matter of minutes took me into the world of panpsychism is where you just convinced me what made me feel like an oil drop is a living certainly an autonomous system but almost the living system so as a capability sensory capabilities and acting capabilities and it maintains something so what is the difference between that and something that we traditionally think of as a living system that it could die or you can't I mean a yab mortality I'm not I'm not exactly sure I'm not sure what the right answer there is because they can move them like movement seems like an essential element to being able to act in the environment but the oil drop is doing that so I don't know easy the mall drop will be moved but does it inner of itself move autonomously well it or the surface is performing actions that maintain its structure yeah you're being too clever I was out of service certified a passive little oil drop this is sitting there yeah and the bottom on the top of a glass sure I guess what I'm trying to say is you're absolutely right you hear you've even nailed it its movement yeah so where does that movement come from if it comes from the inside then then you've got I think something that's living what do you mean from the inside

what I mean is that the internal States the can influence reactive states that where the actor states can influence but they're not influenced by the external states can cause movement so there are two types of oil drops if you like there are oil drops where the internal states are so random that they average themselves away and the thing cannot on a balance on average when you do the averaging move so a nice example of that will be the Sun the Sun Sony has internal States and lots of intrinsic autonomous activity going on but because it's not corded because it doesn't have the deep in the Millennial sense a hierarchical structure and the brain does there is no overall mode or pattern or organization that expresses itself on the surface that allows it to actually swim it it can certainly have you're a very active surface but on mass at the scale of the actual surface of the Sun the average position of that surface cannot in itself move because the internal dynamics are more like a hot gas they are literally like a hot gas whereas your internal dynamics are much more structured and deeply structured and now you can express on your mark of in your active States with your muscles and and your secretary organs your autonomic nervous system and its effectors you can actually move and that's all you can do and that's something which you know if you haven't thought of it like this before I think it's nice just realize there is no other way that you can change the universe other than simply moving whether that moving is articulating my with my voice box or walking around or squeezing juices out of my secreting organs there's only one way you can change the universe it's moving and in the fact that you do so non randomly it makes you alive yeah so it's not non-randomness so that the that's what sohe's and that would be manifesting we realize in terms of essentially swimming essentially moving changing one shape a morphogenesis that is dynamic and possibly adaptive so that that's what I was trying to get out between the difference from the oil drop and the little tadpole the the tampo is moving around its his active states are actually changing the external states and there's now a cycle an action perception cycle if you like a recurrent dynamic that's going on that depends upon this deeply structured autonomous behavior the rests upon internal dynamics that are not only modeling they data impressed upon their surface or the blanket States but they are actively resampling those data by moving they're moving towards Kemet say chemical gradients in chemotaxis so they've gone beyond just being good little models of the kind of world they live in for example an oil droplet could in a pan psychic sense be construed as a little being that has now perfectly inferred it's a passive nonliving oil-drop living in a bowl of water no problem no but to now equipped that oil drop with the ability to go out and test that hypothesis about different states and beings so we can actually push its surface over there over there and test for chemical gradients or then

you start to move to much more lifelike form now this cells is on fun theoretically interesting but it it actually is quite important in terms of reflecting what I have seen since the turn of the millennium which is this move towards it and they inactive an embodied understanding of intelligence and you say you're from machine learning yes so what that means this this sort of the central importance of movement I think has yet to really hit machine learning it certainly has now diffused itself throughout robotics and perhaps you can say certain problems in active vision where you actually have to move the camera to sample this and that but machine learning of the data mining deep learning saw simply hasn't contended with this issue what is done instead of dealing with the movement problem and the active sampling of data it is said we don't need to worry about we can see all the data because we've got big data so we need nor movement so that for me is you know an important omission in current machine learned and current machine learning is much more like the oil drop yes but an oil drop that enjoys exposure to nearly all the data you see to be first as opposed to the tadpoles swimming out to find the right data for example it likes food that's a good hypothesis test analyst go and move and ingest food for example and see what that you know is that evidence that I'm the kind of thing that likes this kind of food so the the next natural question and forgive this question but if we think of sort of even artificial intelligence systems we just paint a beautiful picture of existence and life so do you do you ascribe would you do you find within this framework a possibility of defining consciousness or exploring the idea of consciousness like what you know self-awareness and expand it to consciousness thing yeah how can we how can we start to think about consciousness within this framework is it possible yeah I think it's possible to think about it whether you'll get again I'm not sure that I'm licensed to like question you I think you'd have to speak to a qualified philosopher to get a definitive answer that but certainly there's a lot of interest in using not just these ideas but related ideas from information theory to try and tie down the the maths and the calculus and the geometry of consciousness either in terms of sort of a minimal consciousness III even less than a minimal selfhood and what I'm talking about is the ability effectively to plan so have agency so you could argue that a virus does have a form of agency in virtue of the way that it selectively finds hosts and cells to live in and moves around but you wouldn't endow it with the capacity to think about planning and moving in a purposeful way where it countenances the future whereas you might announce you might think an ants not quite as unconscious as a virus it certainly seems to have a purpose it talks to its friends on route during its foraging it has a different kind of autonomy which is biotic but beyond a virus so there's something about so there's some line

that has to do with the complexity of planning yes that may contain an answer I mean it'd be beautiful if if we can find a line beyond which you could say look being as cautious yes it would be these are wonderful lines that we've drawn with existence life and consciousness yes it will be very nice what one little wrinkle there and this is something I've only learned in the past few months is the notion the philosophical notion of vagueness so you're saying it would be wonderful to draw a line and I had always assumed that that line at some point would be drawn and until about four months ago and the philosopher told me about vagueness so I know if you've come across this but it's a technical concept and I think most revealingly illustrated with at what point does a pile of sand become a pile is it one grain two grains three grains or four grains so at what point would you draw the line between being a pile of sand and a collection of some of the grains of sand in the same way is it right to ask where would I draw the line between conscious and unconscious and it might be a vague concept having said that I agree with you entirely I know systems that have the ability to plan so just technically what that means is your your inferential self evidencing by which I simply mean the dynamics literally the thermodynamics and gradient flows that underwrite the preservation of your oil droplet like form are described as a canvas who has an optimization of Molag Bayesian model evidence in your elbow that self evidencing must be evidence for a model of what's causing the sensory impressions on the sensory part of your surface or your Markov blanket if that model is capable of planning it must include a model of the future consequences who your active States or your action just panning so we're now in the game of planning as inference now notice what we've made though we've made quite a big move away from big data and machine learning because again it's the consequences of moving it's a consequence of selecting those data or those data or looking over there and like that tells you immediately that even to be a contender for a conscious artifact or you know a as it's strong AI or generalize a little no then you've got to have movement in the game and furthermore you've got to have a genitive model of the sort you might find in say a variation or two encoder that is thinking about the future conditioned upon different courses of action now that brings a number of things to the table which which now is start think more those who've got all the right ingredients talk about consciousness I've now got to select an among a number of different courses of action into the future as part of planning I've now got free will the act of selecting this course of action or that policy or that policy or that action suddenly makes me into an inference machine of self evidencing artifact that now looks as if it's selecting amongst different alternative ways forward as I actively swim here or swim there all yes look over here

look over there so I think you've now got to a situation if there is planning in the mix you're now getting much closer to that line if that line whatever to exist I don't think it gets you quite as far as self aware though I think you and then your you have to I think grapple with the question how would formally write down a calculus or a maths of self-awareness I don't think it's impossible to do but I think you would know we pressure on you to actually commit to a formal definition or a mean by self awareness I think most people that I know would probably say that a goldfish no pet fish was not self-aware they would probably argue about their favourite cat but would be quite happy to say that that mom was self-aware so I mean but that might very well connect to some level of complexity with planning it seems like self-awareness is essential for complex planning yeah do you want to talk about further coffee you're absolutely right again the line is unclear but it seems like integrating yourself into the world into Europe into your planning is essential for constructing complex plans yes yeah so I'm mathematically describing that in the same elegant way as you have the free energy principle might be difficult well yes and no I don't think that well perhaps we should just can we just go back that's a very important answer you gave like I think if I just unpacked it you know you'd see the truisms or you've just you've just exposed France but maybe so yeah I'm mindful that I didn't answer your question before well yeah what's the pre hundred principle good for is it just a pretty theoretical exercise to explain non-equilibrium steady states yes it is it does nothing more for you than that it can be regarded as our arrogance but you know it is of the sort of theory of natural selection or a hypothesis of natural selection beautiful undeniably true but tells you absolutely nothing else yeah y-you have legs and eyes and you know it tells you nothing about the actual phenotyping and it wouldn't allow you to build something so the free press or directly the thigh itself is as vacuous as most tautological theories and by tautological of course I'm talking to that you know the theory of Naturalists the survival of the fittest and once the fittest isn't survival while this why cuz the fitter in disc around in circles is in a sense the free energy principle has that same you know deflationary tautology I'm you're under the hood it's yeah it's two ago it's a characteristic of things that exist why they exist because they minimize their energy while the minified know as a free energy because they exist and use keep on going round and round but the what the practical thing which you don't get from natural selection but you could say has now manifest in things like differential evolution or genetic algorithms or NCM see for example in machine learning the practical thing you can get is if it looks as if things that exist are trying to have density dynamics and look as though they're optimizing a variational free energy and a variational free energy has to

be a functional or a gerund to model a probabilistic description of causes and consequences causes out there consequences in the sensorial on the sensory parts of the Markov blanket then it should in theory possible to write down the generative model work out the gradients and then cause it to autonomously self-evidence so you should be able to write down oil droplets you should be able to create artifacts where you have supplied the objective function their supplies the gradients and supplies the self-organizing dynamics to non-equilibrium steady state so there is actually a practical application the free energy principle when you can write down your required evidence in terms of well when you can ride down the Geraldton model that is the thing that has the evidence the probability of these sensory data on this data given that given that model is effectively the thing that the elbow of the variational free energy bounds little proximate s' that means that you can actually write down the model and the kind of thing that you want to engineer the kind of AGI odd artificial general intelligence that you want to manifest probabilistically and then you engineer or world work where you would engineer a robot and a computer to perform a gradient descent on that objective function so it does have a practical implication now why am i Wittering on about that it did seem relevant to yes so what kinds of J so the answer to the but it would it be easier and be hard well mathematically is easy I've just told you all you right down your your perfect artifact probabilistically in the form of a promising charity model probability distribution over the causes and consequences of the world in which this thing is immersed and then you just engineer a computer and a robot to perform a gradient descent on that objective function no problem but of course the big problem is writing down the generative model so that's where the heavy lifting comes in yeah so it's it's the form and the structure of that journal team model which basically defines the artifact that you will create or indeed the kind of artifact that has self-awareness so that's where all the hard work comes very much like natural selection doesn't tell you in the slightest why you have eyes so you have to drill down on the actual feeding time the actual giant ear model so with that in mind what did you tell me that tells me immediately the kinds of journey models I would have to write down in order to have self-awareness what you said to me was I have to have a model that is effectively fit for purpose for this kind of world in which I operate and if I now make the observation that this kind of world is effectively largely populated by other things like me ie you then it makes enormous sense that if I can develop a hypothesis that we are similar kinds of creatures this is in fact the same kind of creature but I am me and you are you then it becomes again mandated to have a sense of self so if I live in a world that is constituted by things like me basically a social world a community then it becomes

necessary now for me to him further it's me talking and not you talking I wouldn't need that if is on Mars by myself or if I was in the jungle as a feral child if there was nothing like it if there's nothing like me around there would be no need to have a an inference that our hypotheses are yes it is me that is experiencing or causing these sounds and it is not you it's only when there's ambiguity and play induced by the fact that there are others in that world so I think then the special thing about self-aware artifacts is that they have learned to or they have acquired or at least are equipped with possibly by evolution Jarett of models that allow for the fact there are lots of copies of things like them around and therefore they have to work out it's you and not me that that's brilliant and I've never thought of that I never thought of that that the purpose of the the really usefulness of consciousness or self-awareness in the context of planning existing in the world is so you can operate with other things like you and like you couldn't it doesn't have to necessarily be human it could be other kind of similar creatures and some absolutely we would view a lot of our attributes into our pets don't waste them or we try to make our robots humanoid and I think there's a your deep reason for that but it's this much easier to read the world if you can make the simplifying assumption that basically you're me and it's just your turn to talk and I mean when we talk about planning when you talk specifically about planning the highest if manifestation or realization of that planning is what we're doing now I mean the human condition doesn't get any higher than this talking about the philosophy of existence and the conversation but in that conversation there is a a you know a beautiful art of turn-taking and it neutral inference theory of mind I have to know when you want to listen I have to know when you want to interrupt and to make sure that you're on line I have to have a model in my head of your model in your head that's the highest the most sophisticated form of generative model where the generative model actually has a generative model somebody else's journeyer model and I think that and what we are doing now evinces the kinds of Geritol models that would support self-awareness because without that we both be talking over each other or we'd be singing together in a in a choir you know we're which is probably not that's not a bridge analogy what I'm trying to say yeah we wouldn't have this discourse yeah what dance of it yeah that's right Dallas to have as I interrupt I mean that's beautifully put I'll really listen to this conversation many times uh there's so much poetry in this and mathematics let me ask the silliest or perhaps the biggest question as the last kind of question we've talked about living in existence and an objective function under which these objects would operate what do you think is the objective function of our existence what's the meaning of life what do you think is the for you perhaps the purpose the source of fulfillment

the source of meaning for your existence as one blob in this soup I'm tempted to answer that as a physicist and free energy I expect consequent upon my behavior so technically that we are and we get a really interesting conversation about what that comprises in terms of searching for information resolving uncertainty about the kind of thing that I am but I suspect that you you you you want a slightly more personal and financial and but which is can be consistent with that and I think it's reassuring is simple and harps back to what you were taught as a child that you have certain beliefs about the kind of creature and the kind of person you are and all that self-evidencing means all that minimizing variational free-energy in an inactive and embodied way means is fulfilling the beliefs about what kind of thing you are and of course we're all given those scripts those narratives very early as usual in the form of bedtime stories or fairy stories I'm a princess I gotta meet a beast who's gonna transform ya a prince and the narratives are all around you from your parents to thee to the friends to the society feeds these stories and then then your objective function is to fulfill exactly now the narrative that has been inculcated by your your immediate family but as you say also the sort of the culture in which you've made you grow and you create for yourself I mean a game because of this active inference it's an inactive aspect of self-evidencing you know not only am I modeling my environment my recognition my my external states out there but I'm actively changing them all the time and it still isn't doing the same back we're doing it together so there's a synchrony that means that I'm creating my own culture over different time scales so the question now is for me being very selfish what scripts were I given it basically was a mixture between an Einstein and a Sherlock Holmes so I smoked as heavily as possible try to avoid too much interpersonal contact yet enjoy the fantasy that the the you know you're a popular scientist who's gonna make a difference and it's like a quirky way so that's what's where I grew up on my father my father was an engineer and loved science and enough he loved real sort of things like Sir Arthur and his space-time and gravitation which was the the other the first understandable version of general relativity and he so although all the fairy stories I was told as I was growing up were all about these characters keeping the Hobbit out of this because I was quite nervous there's a journey of exploration I suppose it's not so yeah I've just grown up to being what I imagine a mild-mannered Sherlock Holmes slush Alvin star would do it in my shoes and you did it elegantly and beautifully Carl was a huge island talking today was fun thank you so much for your time I don't think your shame thank you for listening to this conversation with Carl Wriston and thank you to our presenting sponsor cash app please consider supporting the podcast by downloading cash app and using collect's podcast if you

enjoy this podcast subscribe on youtube review it with five stars an apple podcast supported on patreon are simply connect with me on Twitter and lex friedman and now let me leave you with some words from carl frist in ' your arm moves because you predict it will and your motor system seeks to minimize prediction error thank you for listening and hope to see you next time you